



SEEK-O-SPHERE

README

Project Overview

A free, interactive website that teaches Artificial Intelligence to everyone, ages 8 to 80. Plain HTML, CSS and JavaScript on a Firebase backend.

AI is a tool. You are the thinker. Eulid agrees.

Updated 16 August 2026

What It Is

Seek-O-Sphere teaches Artificial Intelligence in plain language to learners of every age, pairing readable chapters with hands-on practice and personal progress tracking — built with vanilla web technology and Firebase, hosted free on GitHub Pages.

Tech Stack

Layer	Technology
Markup	HTML5 — one single-page app (<code>index.html</code>) holding eleven sections, plus standalone auth and dashboard pages
Styling	Vanilla CSS. Light "Sienna" tokens define the base; <code>css/theme-dark.css</code> overrides them into the dark "Observatory" look the site wears
Scripting	Vanilla JavaScript, no framework
Fonts	Space Grotesk (display) · Inter (body) · Nunito (marquee) · JetBrains Mono (labels)
Questions	Local engine — algorithmic maths and a JSON bank — with an optional Claude API mode
Auth	Firebase Authentication (compat SDK v10.12.0)
Database	Cloud Firestore (compat v10.12.0, offline persistence on)
Hosting	GitHub Pages

What the Site Includes

- **Home** — hero, the alien mascot Eulid, an animated starfield with planets, and a seamless brand marquee.
- **Learn chapters** — What is AI?, History, Types, Study Tools, Using AI Responsibly.
- **History (interactive)** — a 1950→2026 scrubber, a five-era ribbon, a "Then / Now" focus card, landmark chips and a "Guess the Year" quiz.
- **Visual Learning Lab** — Curriculum Lab (CBSE, ICSE, UK, US, IB across Maths, Science, English, History, Geography) and Exam Prep (CUET, Railways, Bank PO, SSC, UPSC), both with the Eulid tutor, hints, explanations, streaks and saved progress.
- **Find My AI Tool** — a keyword-scored recommender over 20+ tools.
- **Help Centre** — a searchable FAQ with a calm topic-card landing.
- **Study planner** — a side panel with day, week and month views.
- **Accounts** — email or Google sign-in, plus guest browsing.
- **Student Dashboard** — accuracy, best streak and active days, a progress chart, per-subject breakdown, an activity heatmap and recent sessions.

- **Admin Panel** — a user list with scores and a per-user progress viewer. School admins see only their own school.

Eulid, the Mascot

Eulid is a small feature in his own right, living in `js/components/eulid.js`, `css/components/eulid.css` and the shared markup in `assets/eulid.html`.

- **There is only ever one of him.** On the home page the element that floats is literally the one from the hero, moved; elsewhere the markup is fetched once. This matters because he is drawn from three gradients addressed by `id`, so a second copy would silently take the first one's fills.
- **Hold and drag** him anywhere. A 4px threshold separates a press from a drag, so pressing a leg is never mistaken for moving him.
- **His five legs are tools** — magnifier, multi-colour highlighter with an eraser, read-aloud, a dashboard shortcut, and one the user defines. Esc or a second press turns any of them off.
- **He follows you, then goes home.** His position lives in `sessionStorage` with a timestamp refreshed while the page is visible, so he travels with you and returns to his default place after half an hour away or when the tab closes.

Project Structure

```
ai-classroom/
├ index.html           the app: eleven sections shown via showPage()
├ README.md  BACKEND_SETUP.md  HANDOFF.md
├ firestore.rules      security rules (publish in the Firebase console)
├ structure.js         page-shape regression check
├ docs/               these PDFs
├ pages/              login, dashboard, admin, account-settings, mission,
│                     plus thin nav shells for the content sections
├ css/                theme-dark.css, dashboard.css, base/, components/,
│                     layout/, pages/
├ js/                 session.js, questions-engine.js, core/, components/,
│                     auth/, db/, pages/, tools/
├ data/               question-bank.json, tools-data.js
└ assets/             eulid.html, backdrop.html, images/, video/
```

Two faces. `index.html` is the canonical experience — home, what, history, types, study, ethics, lab, finder, resources, help and contact all live inside it and are swapped by `showPage()`. Most files under `pages/` are thin shells that mount the shared nav and footer; the substantial ones are login, dashboard, admin, account-settings and mission.

Design System

The site ships two token sets. The light Sienna values are the defaults in `:root`; `theme-dark.css` restates the same names under `html[data-theme="dark"]`, which every page sets. Because the site paints through tokens, restating twenty-odd values converts roughly a thousand declarations at once.

--obs-void #05080E	The ground everything sits on
--obs-panel #0C121C	Raised surfaces: cards, bars
--obs-ink #E6EDF6	Body text
--obs-dim #8496AB	Secondary text — 6.6:1 on the void
--obs-aqua #63E6E2	The one signal colour
--obs-ember #FF8A4C	Held back for what matters most

Each typeface has one job: **Space Grotesk** for every display role, **Inter** for anything meant to be read, **Nunito** for the scrolling marquee only, **JetBrains Mono** for small uppercase labels. The quote scroller — Space Grotesk 400 in `#E6EDF6` — is the site's "someone is speaking" treatment, and anything that speaks should join it rather than invent a new style.

Local Development & Deployment

The site is fully static: open it with any static server, no build step and no `npm install`. Work happens on `dev`; deployment is a merge into `main`, which GitHub Pages serves.

GitHub Pages serves static files only — it does **not** apply `firestore.rules`. Publish those in the Firebase console. See `BACKEND_SETUP`.